Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**A Physics-Informed Sim2Real Foundation Model for Autonomous Assembly**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical · Materials			
	Medium Category	*Small Category*	*Medium Category*	*Small Category*
	Robotics/Automation Machinery	*Robot Control and Intelligence Technologies*		

Concept of Proposed Technology

Core Concept

This project aims to bridge the "Sim-to-Real Gap" in manufacturing automation by integrating software-based Physical Foundation Model with real-world robot control and verification platform (JARVIS). The goal is to develop a VLA (Vision-Language-Action) robot intelligence capable of autonomously assembling deformable objects (e.g., cables, rubber, EV battery modules) without human expertise.

Key Functions

1) Physics-Informed Foundation Model
   - Physics-Informed Neural Networks (PINN): Incorporating physical laws (Newtonian mechanics, material properties) into the AI learning process to ensure the model adheres to physical constraints, reducing data requirements for training
   - High-Fidelity Simulation: Creating a "Ground-Truth" virtual environment that

perfectly models the non-linear behavior of deformable materials, enabling the generation of massive datasets including rare "corner cases"

2) Real-world Deployment & Control

- Zero-Shot Sim2Real Transfer Engine: Utilizing Residual Reinforcement Learning to correct simulation errors with minimal real-world data (50~100 shots), achieving a 97% match rate.
- Force-Aware VLA Controller: A real-time controller integrating Vision-Language-Action models with 6-axis Force/Torque sensors. It provides reflexive control (<15ms latency) to handle contact-rich tasks like "snap-fits" or cable insertion

Potential Applications

Autonomous assembly for EV battery modules (busbars, FPCBs), cable harnesses, and flexible rubber parts (gaskets, tires)

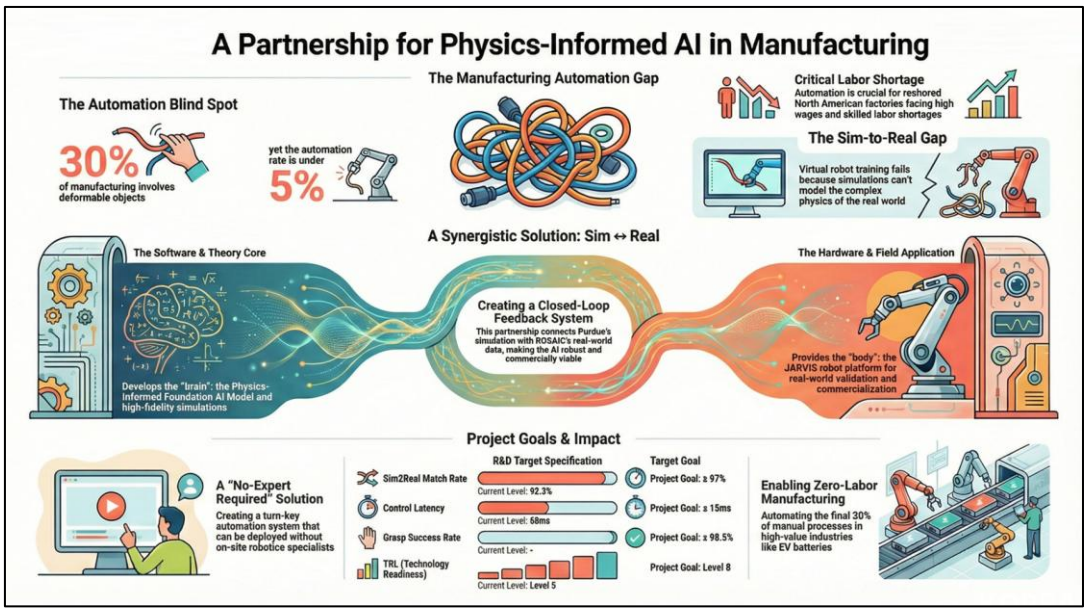

Figure. Target Application & Engineering Rationale

Rationale for Joint R&D Project

Critical Technical Barriers

Deformable objects exhibit non-linear dynamics and complex contact behaviors that traditional coordinate-based robots cannot model, making precise assembly extremely difficult. This limitation creates a significant Sim-to-Real Gap—the discrepancy between simulation and real-world factory environments—which leads to frequent failures in precision assembly and results in automation rates below 10% in these sectors. At the same time, domestic technologies lack the ground-truth physics models required for robust AI generalization. Conventional data-driven and black-box AI approaches fail to handle the “infinite variability” of deformable objects in real-world factories without precise physical modeling, and are unable to reliably address rare but critical corner cases encountered in practical manufacturing environments.

Why Global Collaboration?

World-class capabilities in physics-based simulation and foundation model architectures are

	essential to build the “Brain” (software/models) that can generate high-fidelity training data and robust control policies—capabilities that cannot be achieved through data collection alone. At the same time, hardware integration and real-world field validation are required to verify and refine these models under practical manufacturing conditions. By combining advanced software/model expertise with hardware and field validation capabilities, this project establishes a closed-loop feedback system (Sim ↔ Real), ensuring that the technology moves beyond theory to become commercially viable for factories facing skilled labor shortages. **Expected Impact** By internalizing theoretical physics-based AI, the project enables the acquisition of sovereign technology for handling deformable objects, reducing dependence on foreign-owned automation software. At the same time, it strengthens the domestic manufacturing ecosystem by providing low-cost, high-efficiency automation solutions for small and medium-sized enterprises struggling with labor shortages.
Key Words	1. Physical Foundation Model 2. Sim2Real Transfer 3. VLA (Vision-Language-Action) 4. Deformable Object Manipulation 5. Physics-Informed AI

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	**R&D Objectives** Develop a commercial-grade Physics-Informed Sim2Real Foundation Model and "Turn-key" Robot Cell capable of autonomous assembly of deformable objects. **Detailed Objectives** 1) 1. Physics-Informed Foundation Model ● Develop an AI model that understands non-linear material dynamics. 2) 2. Sim2Real Transfer Engine (Joint) ● Achieve >97% simulation-to-reality policy match rate. 3) 3. Industrial Validation ● Validate reliability in real-world manufacturing lines (e.g., battery module assembly) to ensure industrial reliability.			
R&D Target Specifications	1) Sim2Real Match Rate ≥ 97%, Weight: 30%, RMSE comparison of simulation vs. real sensor trajectories 2) Control Latency ≤ 15ms, Weight: 20%, End-to-end response time measurement (Oscilloscope) 3) Grasp Success Rate ≥ 98.5%, Weight: 20%, Success rate over 1,000 trials with deformable objects 4) Task Setup Time < 24 Hours, Weight: 15%, Time from installation to full operation (Zero-shot) 5) Corner Case Recovery≥ 92%, Weight: 15% Autonomous recovery rate from induced errors (500 cases			
R&D Plan	Year 1: Foundation Model Development & Simulation Setup ● Develop Physical Foundation Model architecture and high-fidelity simulation environment for deformable objects. ● Define manufacturing use-cases and collect initial real-world baseline data. Year 2: Sim2Real Transfer & System Integration ● Implement Sim2Real transfer algorithms (Residual RL) to bridge the reality gap. ● Integrate the model into the JARVIS-CTRL hardware and conduct lab-scale verification. Year 3: Field Validation & Commercial Optimization ● Deploy the system to actual manufacturing lines (e.g., EV battery assembly) for stress testing. ● Finalize the "Turn-key" solution for the North American market and secure reliability certification. **Methodology** 1. Physics-Informed Learning			

	● Embedding differential equations of material dynamics into the loss function of the neural network to ensure physical plausibility. 2. Domain Randomization ● Randomizing simulation parameters (friction, light, mass) to cover the "Reality Gap." 3. Edge-AI Optimization ● Utilizing NVIDIA TensorRT's layer fusion and kernel auto-tuning to achieve end-to-end inference latency under 15ms, enabling real-time reaction to material deformation.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Software & Theory Core: The "Brain"** Provide source technology for Physics-Informed AI and Multi-physics Simulation of deformable materials (e.g., rubber, cables). 1. Develop Physics-Informed Foundation Models and learning algorithms for deformable object manipulation. 2. Design high-fidelity simulation environments to generate massive training data covering corner cases. 3. Establish benchmarking protocols for Sim2Real transfer performance.
Expected R&R of Korean SME Partner	**Hardware & Field Application: The "Body"** Develop Real-time VLA Controllers and commercialize the "Turn-key" Robot Cell (JARVIS-Cell). 1. Develop and optimize the real-time robot controller (HW) and safety layers for industrial deployment. 2. Conduct field validation (PoC) in manufacturing plants to collect real-world feedback data. 3. Commercialize the solution as a "Turn-key AX Robot Cell" for global market expansion.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**(Technical) Global Standard for Physical AI** Establish a Physics-Informed Sim2Real framework that overcomes the limitations of data-driven AI, solving the "Black-box" problem of current AI robots. Achieve 97% reality match, setting a new global benchmark. Establish a global standard for Physical AI in manufacturing, overcoming the limitations of traditional coordinate-based robots and enabling the automation of complex deformable tasks. **(Economic) ROI Maximization for Reshoring Factories** Provide a "No-Expert Required" solution aligned with North American reshoring trends. Eliminating the need for system integrators reduces setup costs by 80% and setup time from weeks to <24 hours, offering immediate ROI for high-wage markets. Provide a "No-Expert Required" automation solution perfectly aligned with the reshoring trends in North America, addressing the severe shortage of skilled labor and high wages. **(Industrial) Solving the Skilled Labor Crisis** Enable Zero-Labor manufacturing for the 30% of processes (deformable parts) that are currently manual. This directly addresses the aging workforce and labor shortage issues in advanced manufacturing nations. Achieve Zero-Labor manufacturing for high-value industries (EV batteries, semiconductors) by automating the remaining 30% of manual processes.

6. References

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